OKBA BOUZIANE

Full-Stack & AI Engineer

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TECHNICAL SKILLS

AI & LLM: LangGraph, LangChain, LlamaIndex, AI Agents, RAG, Graph RAG, MCP, PGVector, Prompt Engineering, LLM APIs
Frontend: Angular, React, TypeScript, RxJS, React Native
Backend: Node.js, NestJS, Express.js, Python, FastAPI, REST, GraphQL, gRPC
Databases: PostgreSQL, MongoDB, Neo4j, Redis, MinIO (S3)
Security & DevOps: Keycloak, OAuth2 / OpenID Connect, RBAC, Docker, CI/CD, Git
AI Tools: Claude Code, Cursor, Windsurf, Kiro

PERSONAL PROJECTS

Secure Agent Workspace | React, NestJS, LangGraph, Keycloak, PostgreSQL
An AI assistant that works on company data with the permissions of the user who is logged in.

- The agent can only do what the user is allowed to do, and every change waits for a human to approve it.
- Answers questions from company documents with sources, blocks prompt injection, and has 82 automated tests.

VoiceDesk | Python, FastAPI, LiveKit, LangGraph, gRPC, PostgreSQL
A voice receptionist you talk to in the browser, live online.

- Books, lists and cancels appointments by voice, and answers questions from a knowledge base.
- Hands the call to a human when needed; bookings run in a separate gRPC microservice.

LexGraph | Python, FastAPI, LlamaIndex, Neo4j, LangGraph
A contract search assistant that follows the links between documents.

- Stores contracts, amendments and guarantees as a Neo4j knowledge graph to answer multi-document questions.
- On those questions it finds 92% of the needed documents versus 74% for plain vector search; covered by 41 automated tests.

PROFESSIONAL EXPERIENCE

Full-Stack Developer | Smartest Algeria, Algiers — Nov 2022 – Oct 2026

- On a well-integrity platform for oil & gas operators, corrosion risk was assessed well by well, missing failures that spread across neighbouring wells; built a LangGraph agent that evaluates each well against the inspection history and corrosion status of nearby wells, using live geospatial and integrity data as tools, and flags at-risk wells for engineers to confirm.
- As one of 4 backend developers, built Node.js microservices for well-data ingestion, integrity assessment and reference data; audited the bug-prone Angular screens, fixed the defects and rebuilt them on the shared component library.
- Operations teams reviewed documents and filled forms by hand; integrated LLM APIs for document summarization, smart search and form auto-fill, built end to end with Angular, React and NestJS.
- Each product had its own login and access rules; integrated Keycloak (OAuth2/OpenID Connect) with JWT-secured APIs and RBAC for single sign-on across the apps, and built an admin console on the Keycloak Admin REST API.
- Licensing was handled separately in each app; built a SaaS licensing platform that centralizes it for every Smartest app.
- Business users relied on developers for every new report; built a Word-like report designer where they bind datasets to tables and charts and generate reports themselves.
- Several apps needed visual flow editing and would each have rebuilt graph rendering; designed flow-dcl, a reusable Angular flow-graph editor library with an extension API for custom node and edge types.
- Optimized PostgreSQL queries behind slow reports, added MinIO (S3) file storage, and shipped with Docker and CI/CD.

Mobile & Web Developer | Tebessa Innoest, Tebessa — Mar 2020 – Jan 2022

- Incubated founders had ideas but no product to pitch; turned them into mobile and web MVPs in short delivery cycles, advising on technical scope and feature priority so each team could demo a working first version.
- Start-ups needed apps on both platforms on a small budget; built cross-platform React Native apps (iOS and Android) with authentication, REST APIs, push notifications and offline-friendly screens, plus React websites for demo days.

EDUCATION

MSc, Artificial Intelligence | Badji Mokhtar University, Annaba — 2020 – 2022
BSc, Computer Science | Badji Mokhtar University, Annaba — 2017 – 2020

ADDITIONAL INFORMATION

Languages: Arabic (Native), English (Full Professional), French (Limited Working) | **Military Service:** Exempt